ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

NAME ENTITY RECOGNITION

UC San Diego

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Classic Named Entity Recognition (NER)

Some questions to think about:

- Can massive raw texts help?
- Can dictionaries help?
- Are human annotations always correct?
- Is Tokenizer always good?

WHAT'S NAMED ENTITY RECOGNITION?

Wikipedia:

- **Named-entity recognition (NER)** is a subtask of **information extraction (IE)** that seeks to **locate** and **classify named entities** in text into **pre-defined categories**.
- In IE, A **named entity** is a real-world object.

Example

- Input:
  - Jim bought 300 shares of Acme Corp. in 2006.
- Output:
  - $[\text{Jim}]_{\text{Person}}$ bought 300 shares of $[\text{Acme Corp.}]_{\text{Organization}}$ in $[\text{2006}]_{\text{Time}}$.

SUPERVISED METHODS: TRAINING DATA

Sequence labeling framework

Two popular schemes

- BIO: **B**egin, **I**n, **O**ut
- BIOES: **B**egin, **I**n, **O**ut, **E**nd, **S**ingleton
- BIOES is arguably better than BIO (Ratinov and Roth, ACL 09)

Example:

- LABELS: $[\text{Jim}]_{\text{Person}}$ bought 300 shares of $[\text{Acme Corp.}]_{\text{Organization}}$ in $[\text{2006}]_{\text{Time}}$.
- TOKNES: Jim bought 300 shares of Acme Corp. in 2006 .
- BIO: B-PER O O O O B-ORG I-ORG O B-Time O
- BIOES: S-PER O O O O B-ORG E-ORG O S-Time O

Handcrafted Features + Conditional Random Fields

Popular Tools

- Stanford NER (Finkel et al., ACL'05)
  - https://nlp.stanford.edu/software/CRF-NER.shtml
- Ohio State University Twitter NER (Ritter et al., EMNLP'11, KDD'12)
  - https://github.com/aritter/twitter_nlp

CLASSICAL SUPERVISED METHODS: FEATURE ENGINEERING

Bag-of-words Features

- Unigrams, Bigrams, …
- Lowercase
- Lemmazation
- Stemming
- Part-of-speech Tags
- Word Shape

Context Features

- E.g., Left/right 4 words

Character n-grams

- Useful for unknown words

Brown Clusters

- Data-driven similar words
- Alleviate sparsity issues

WordNet Clusters

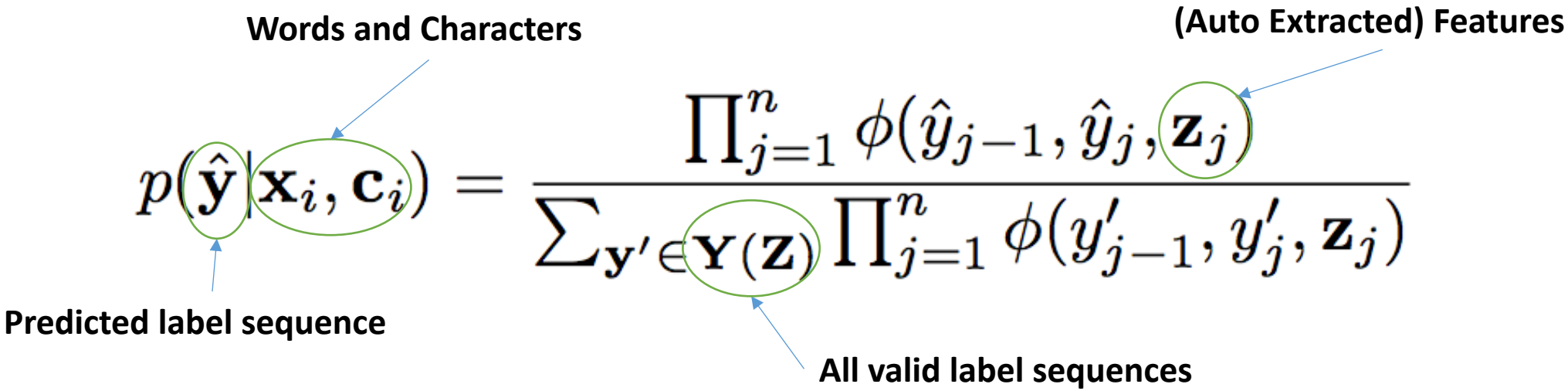
- Semanticly similar words

Dictionaries

- Domain-specific dictionaries

Conditional Random Field (CRF) model

- Capture the dependencies between labels



CLASSICAL SUPERVISED METHODS: DRAWBACKS

They require:

- Human Expert Efforts = Domain + Linguistic
- Every single sentence must be fully annotated

Therefore, it is difficult to adapt them to new domain from the original training domain

- New training data → Expensive in specific domains (e.g., biomedical)

Two pioneer models

- LSTM-CRF (Lample et al., NAACL'16)
- LSTM-CNN-CRF (Ma and Hovy, ACL'16)

	LSTM-CRF	LSTM-CNN-CRF
Word-Level	Bidirectional LSTMs	Bidirection LSTMs
Character-Level	Bidirectional LSTMs	Convolutional NN

- The first neural model that outperforms the models based on handcrafted features

LET'S TRY SPACY FOR NER TOGETHER

In [1]:
```
import spacy

nlp = spacy.load('en_core_web_sm')
```

In [2]:
```
sentence = "Apple is looking at buying U.K. startup for $1 billion"

doc = nlp(sentence)

for ent in doc.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
    print('\t', sentence[ent.start_char : ent.end_char]) # the start_char and end_char are offs
```

```
Apple 0 5 ORG
	 Apple
U.K. 27 31 GPE
	 U.K.
$1 billion 44 54 MONEY
	 $1 billion
```

LET'S TRY SPACY FOR NER TOGETHER

Capitalization

Some old version of spaCy is sensitive to capitalization but the new version now is using deep neural network models, which is much better.

```
spaCy v2.0's Named Entity Recognition system features a sophisticated word embedding st
rategy using subword features and "Bloom" embeddings, a deep convolutional neural netwo
rk with residual connections, and a novel transition-based approach to named entity par
sing.
```

In [3]:
```
sentence = "apple is looking at buying U.K. startup for $1 billion"

doc = nlp(sentence)

for ent in doc.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

```
U.K. 27 31 GPE
$1 billion 44 54 MONEY
```